

WASP-43b

R-2712

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-43_130130 · created 2026-09-01T01:15:04+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2456322.82015 +/- 0.00052 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.147 +/- 0.018
Transit depth (Rp/Rs)^2	2.15 +/- 0.54 [%]
Orbital Inclination (inc)	81.69 +/- 2.03
Airmass coefficient 1 (a1)	0.935 +/- 0.012
Airmass coefficient 2 (a2)	0.045 +/- 0.01
Scatter in the residuals of the lightcurve fit is	1.31 %
Best Comparison Star	#2 - [272, 305]
Optimal Method	PSF photometry
Transit Duration (day)	0.0471 +/- 0.0082

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R* fit vs published	0.147 ± 0.018 vs 0.1594 (deviation 0.69σ)
Duration fit vs published	0.0471 d vs 0.0512 d (deviation 0.51σ)
Mid-time O-C	1.0 min
Depth SNR	8.2
Residual scatter	1.31 %

Light curve

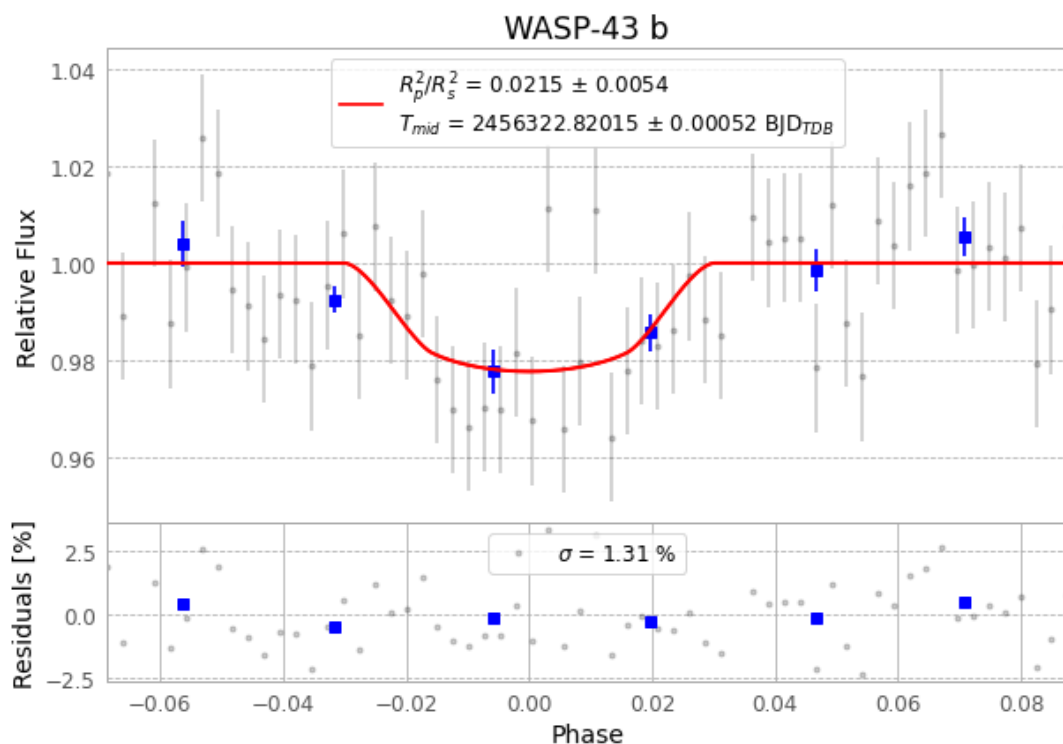


Figure 1 — FinalLightCurve_WASP-43 b_2013-01-29.png (SHA-256 48d5c0059c922bb8...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [186, 258], 3 comparison star(s). Exact configuration: parameters.inits.json in record.json (SHA-256 bb88d4927a05ceeb...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit a4253ce

Data provenance

62 FITS frames (41 MB), WASP-43130130061212.FITS → WASP-43130130091512.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-43-130130.log (95483bedfca2...), FinalLightCurve_WASP-43 b_2013-01-29.png (48d5c0059c92...), FinalLightCurve_WASP-43 b_2013-01-29.csv (f8c2be5f6e07...)

Compute resources

Wall time 102.8 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-09-01 01:15Z.